WORKED SOLUTION

STEP III 2015, Question 9

Mechanics • Elastic string

STEP III 2015, Question 9

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/15-s3-q9

The key idea

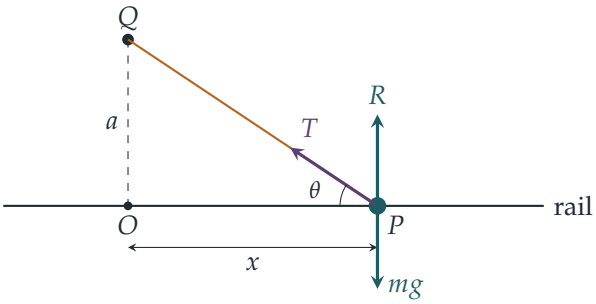
The rail is smooth and the normal reaction does no work, so energy is conserved, and that gives the first-order equation straight away. The string is never slack, because $PQ \geq a$ at every point of the rail. Turning points are where $\dot{x} = 0$. The period is four times the time to go from the centre out to x_0. For small v the particle stays close to its start, so $\sqrt{x^2 + a^2} - a \approx x^2/(2a)$ and the substitution $x = x_0 u$ tidies up the integral.

Solution

The energy equation

Model P as a particle and the string as light and elastic, obeying Hooke's law. Take the rail as the x-axis with P starting at the origin, the foot of the perpendicular from Q. When P is at x, its distance from Q is $PQ = \sqrt{x^2 + a^2} \geq a$, so the string is taut, with extension

$$e = \sqrt{x^2 + a^2} - a \quad (\geq 0).$$



The forces on P are its weight, the normal reaction R of the rail (both perpendicular to the motion, so they do no work) and the tension T. The tension is a conservative force with elastic potential energy $\lambda e^2/(2a)$. At the start $e = 0$ and the speed is v, so

conservation of energy gives

$$\tfrac{1}{2}m\dot{x}^2 + \frac{\lambda}{2a}\Big(\sqrt{x^2+a^2} - a\Big)^2 = \tfrac{1}{2}mv^2.$$

Multiplying by $2/m$ and writing $k^2 = \lambda/(ma)$,

$$\dot{x}^2 = v^2 - k^2\Big(\sqrt{x^2+a^2} - a\Big)^2,$$

as required.

The greatest displacement and the acceleration there

The extension e increases with $|x|$, so $\dot{x}^2$ decreases as P moves out, and P moves out until $\dot{x} = 0$. At that point $k\big(\sqrt{x_0^2+a^2} - a\big) = v$ (taking the positive root, since $e \geq 0$). So

$$\sqrt{x_0^2+a^2} = a + \frac{v}{k} \quad\Longrightarrow\quad x_0^2 = \Big(a + \frac{v}{k}\Big)^2 - a^2 = \frac{2av}{k} + \frac{v^2}{k^2},$$

giving

$$x_0 = \frac{\sqrt{v\,(2ak+v)}}{k}.$$

For the acceleration, resolve along the rail. If θ is the angle between PQ and the rail, then $\cos\theta = x/\sqrt{x^2+a^2}$, and the component of the tension along the rail is $T\cos\theta$ towards O. With $T = \lambda e/a$,

$$m\ddot{x} = -\frac{\lambda}{a}\Big(\sqrt{x^2+a^2} - a\Big)\frac{x}{\sqrt{x^2+a^2}}, \qquad \text{so} \qquad \ddot{x} = -k^2\Big(\sqrt{x^2+a^2} - a\Big)\frac{x}{\sqrt{x^2+a^2}}.$$

(Differentiating the energy equation with respect to t and cancelling $2\dot{x}$ gives the same thing.) At $x = x_0$ we have $\sqrt{x_0^2+a^2} - a = v/k$ and $\sqrt{x_0^2+a^2} = (ak+v)/k$, so

$$\ddot{x} = -k^2 \cdot \frac{v}{k} \cdot \frac{k\,x_0}{ak+v} = -\frac{k^2 v\,x_0}{ak+v} = -\frac{kv\sqrt{v\,(2ak+v)}}{ak+v}.$$

So the acceleration at x_0 has magnitude $\dfrac{kv\sqrt{v(2ak+v)}}{ak+v}$ and is directed back along the rail towards the initial position.

The period

The equation for $\dot{x}^2$ depends only on x^2, so the motion is symmetric about O. The particle oscillates between $-x_0$ and x_0, and each of the four quarters $O \to x_0 \to O \to -x_0 \to O$ takes the same time. On the first quarter $\dot{x} > 0$, so $\mathrm{d}t = \mathrm{d}x/\dot{x}$ and the period is

$$\tau = 4\int_0^{x_0} \frac{\mathrm{d}x}{\sqrt{v^2 - k^2\big(\sqrt{x^2+a^2} - a\big)^2}}.$$

The integrand blows up at x_0, but only like $(x_0 - x)^{-1/2}$, so the integral converges.

Ask yourself What does $v \ll ka$ tell us about the size of x?

Since $x_0^2 = 2av/k + v^2/k^2$, if $v/k \ll a$ then $x_0^2 \approx 2av/k$, so $x_0^2/a^2 \approx 2v/(ka) \ll 1$. Every x in the range of integration therefore satisfies $|x| \ll a$, and the binomial expansion gives

$$\sqrt{x^2+a^2} - a = a\left(1 + \frac{x^2}{a^2}\right)^{1/2} - a \approx \frac{x^2}{2a}.$$

The period becomes

$$\tau \approx 4\int_0^{x_0} \frac{\mathrm{d}x}{\sqrt{v^2 - k^2x^4/(4a^2)}}, \qquad x_0 \approx \sqrt{\frac{2av}{k}},$$

and this approximate x_0 is exactly where the new denominator vanishes, since $k^2x_0^4/(4a^2) = v^2$. Now substitute $x = x_0u$, so that $\mathrm{d}x = x_0\,\mathrm{d}u$ and $k^2x^4/(4a^2) = v^2u^4$:

$$\tau \approx 4\int_0^1 \frac{x_0\,\mathrm{d}u}{v\sqrt{1-u^4}} = \frac{4}{v}\sqrt{\frac{2av}{k}}\int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^4}} = \sqrt{\frac{32a}{kv}}\int_0^1 \frac{\mathrm{d}u}{\sqrt{1-u^4}},$$

since $\frac{16}{v^2}\cdot\frac{2av}{k} = \frac{32a}{kv}$. This is the required approximation.

Checks

- Units: $k^2 = \lambda/(ma)$ has units of s^{-2}, so v/k is a length (as it must be, since it is the extension at x_0) and $\sqrt{a/(kv)}$ is a time.
- With $m = 1$, $\lambda = 2$, $a = 1$, $v = 1$ we get $k = \sqrt{2}$, $x_0 = \sqrt{1+2\sqrt{2}} \approx 1.3836$ and acceleration ≈ -1.1462 at x_0. Stepping the equation of motion forward numerically gives the same turning point, and a period of about 7.612, which matches the integral.
- For $a = 1$, $k = 2$, $v = 0.001$, the exact period integral is 165.862, against 165.833 from the approximation, where $\int_0^1 (1-u^4)^{-1/2}\,\mathrm{d}u \approx 1.3110$. The ratio tends to 1 as $v \to 0$.

Common slips

- Using the extension x or $\sqrt{x^2+a^2}$ instead of $\sqrt{x^2+a^2} - a$; the string starts at its natural length.
- Forgetting to resolve the tension: only the component $T\cos\theta$ acts along the rail.
- Taking the period as $2\int_0^{x_0}$ or $\int_0^{x_0}$; the oscillation is between $-x_0$ and x_0, so four quarter-periods are needed.

Beyond STEP

For small v the restoring force is roughly proportional to x^3, not x, so this is a nonlinear oscillator whose period depends on its amplitude, growing like $v^{-1/2}$ as the energy shrinks, unlike simple harmonic motion. The integral $\int_0^1 (1-u^4)^{-1/2}\,\mathrm{d}u$ is an elliptic integral (the lemniscate constant divided by two), the kind that appears for the exact period of a pendulum too. Kleppner and Kolenkow, *An Introduction to Mechanics*, covers energy methods and oscillations in this spirit.

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